



On production planning and scheduling in food processing industry: Modelling non-triangular setups and product decay



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ABSTRACT

Production planning and scheduling in food processing industry (FPI) requires taking specific characteristics into account. First of all, setups are usually sequence-dependent and may include the so-called non-triangular setup conditions. Secondly, planning problems in FPI must take product decay into consideration. We present an MILP model that handles these characteristics. We study its behaviour and complexity and show that optimal production schedules become significantly different when non-triangular setups and product decay are taken into account. Numerical results are provided for medium size instances, including a comparison with a standard MP-based heuristic.

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1. Introduction

Adequate and efficient production planning and scheduling is one of the most challenging problems for present-days enterprises. Lot-sizing and scheduling in Food Processing Industry (FPI) is usually more complex than in other continuous and discrete processing environments. First of all, planners have to deal with decline in quality of products, related quality requirements and safety regulations of products, market driven standards regarding shelf life, and variability of demand and prices. Secondly, the diversity of products in FPI increased considerably in the past decades and global competition on the food market has forced manufacturers to participate in an on-going trend towards increased variety (e.g. ingredients and flavours, customised packaging, prints and/or labels) of (new) products. Soman et al. [54] state that the majority of research contributions do not address specific characteristics of food processing, e.g. high capacity utilisation, sequence-dependent setups and limited shelf life due to product decay.

In general practice, lot-sizing and scheduling problems are solved separately in successive hierarchical phases [12,17,40,54,56]. First optimal lot-sizes for given product families are determined and afterwards production schedules are generated. The generated schedules on the shop floor often fail to realise production targets, because changeover losses are not correctly accounted for on a

higher planning level. As a consequence, the planning process has to be redone (with or without over-time) and/or frequent re-scheduling takes place in daily practice [40]. Currently, there exists a general consensus regarding a closer integration of lot-sizing and scheduling decisions, see Meyr [44], Gupta and Magnusson [33], Jans and Degraeve [36], Almada-Lobo et al. [2], Clark et al. [13], and Menezes et al. [43]. Guimarães et al. [31] survey the main modelling approaches and present a classification framework for integrated lot-sizing and scheduling models. Some recent real-life applications of simultaneous lot-sizing and scheduling can be found for instance in [24,7,9]. In spite of the increasing attention for simultaneous lot-sizing and scheduling in literature, surprisingly little research has been done to include specific aspects of food industry (e.g. product decay) into traditional lot-sizing and scheduling models.

Planning (i.e. lot-sizing) models differ from scheduling models in a number of ways. Kreipl and Pinedo [40] give an extensive overview of practical issues for planning and scheduling processes. In a special issue on lot-sizing and scheduling, Clark et al. [13] confirm the need for more realistic and practical variants of models for simultaneous lot-sizing and scheduling. Features such as (i) non-triangular setups, (ii) perishability, and (iii) delivery time windows are labelled by the authors as hot topics and open research opportunities. The research question of this paper is how to include the first two characteristics in models for simultaneous lot-sizing and scheduling.

- (1) *Sequence-dependent setups and non-triangular setups.* There is a complicating issue with respect to sequence-dependent

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setup costs and times, commonly referred to as the assumption of the triangular setup conditions [13,14,2,33]. Menezes et al. [43] confirm that non-triangular setups may occur in FPI due to contamination between production lots. Clark et al. [14] mention that contamination is a particularly concern for FPI. The authors refer for instance to the severe (i.e. lethal) impact of copper in concentrates for sheep due to an ineffective sequencing of production lots. Due to processing conditions of different product variants (e.g. several heating and/or cooling levels) and other product specific requirements (e.g. flavours, addition of specific additives, the danger of contamination between subsequent production runs), changeover costs and times between two subsequent products i and j may become substantially less by processing another product k between i and j . As a consequence, applying models that assume triangular setup conditions may generate non-consistent solutions from a scheduling point of view.

- (2) *Product decay*. In many FPI cases, the quality or value of perishable food products deteriorates rapidly after production. Considering product decay in lot-sizing enforces smaller production quantities. Consequently, individual products are produced at higher frequency. This increases the difficulty of sequencing.

This paper investigates implementing the characteristics into models for simultaneous lot-sizing and scheduling under tight capacity constraints. We present an MILP model that includes the identified characteristics. Moreover, the approach offers a natural starting point for integrating delivery time windows in lot-sizing and scheduling models as mentioned by Clark et al. [13]. Small scale examples demonstrate that optimal production schedules become significantly different when including non-triangular setups and product decay. Two model formulations are presented and compared with a known approach from literature.

The remainder of the paper is organised as follows. [Section 2](#) embeds the model in existing approaches from literature. [Section 3](#) presents two MILP models for the problem under consideration. [Section 4](#) provides small scale numerical examples to demonstrate the impact of non-triangular setups and product decay. Moreover, the complexity of the model is studied. [Section 5](#) provides numerical results for medium size instances, including a comparison with a straightforward MP-based heuristic. Concluding remarks and suggestions for further research are given in [Section 6](#).

2. Embedding in the literature

Models for lot-sizing and scheduling can be classified according to the segmentation of the planning horizon. From a modelling point of view, it is convenient to distinguish two general classes of models [19], i.e. small bucket (SB) and big (or large) bucket (BB) modelling approaches. In SB models, the planning horizon is divided into a finite number of small time periods such that in each period either at most two products can be produced, or there will be no production at all. Conversely, in BB approaches the planning horizon is divided into longer periods, usually of equal length. In each period, multiple products may be produced. As a consequence, SB models are usually associated with short term planning horizons and BB models with medium term planning horizons.

2.1. Small bucket approaches

A typical example of SB approaches is the Discrete Lot-sizing and Scheduling Problem (DLSP). The basic DLSP includes (sequence-independent) setup costs and setup carry-over at zero

setup time [25]. Inclusion of setup carry-over implies that setup states of a machine are carried over between period boundaries. Porkka et al. [49] compare models with and without setup carry-overs. They show that substantial savings in costs and production time can be achieved by fundamentally different production plans enforced by carry-overs. Comparable results are found by Sox and Gao [57]. However, in the basic DLSP, setup states are not preserved over idle time. Sequence-dependent setup costs and times are neither considered in the DLSP. Many extensions of the (basic) DLSP have been described in literature. We refer to Drexel and Kimms [17] and Salomon et al. [52] for a broader view on variants of the DLSP.

Fleischmann [26] analyses the multi-item single machine DLSP with sequence-dependent setup costs. An artificial product is introduced to represent idleness of the machine. Salomon et al. [53] continue the latter study and reformulate a DLSP that captures sequence-dependent setup times (DLSPSD). The triangular setup conditions are assumed to hold. However, machine idleness is represented by an artificial product. Jordan and Drexel [37] present a comparable model in which idleness is indicated by an artificial product too. It should be mentioned that if idleness is represented by an artificial product, the changeover matrix must fulfil very strict conditions to cope with sequence-dependent setup times. Otherwise the setup state of the machine is not correctly carried over across the boundaries of idleness. A recent approach is due to Guimarães et al. [31]. The authors propose a classification framework to survey and classify the main modelling approaches for the integration of sequencing decisions in discrete time lot-sizing and scheduling models.

Wolsey [62] extended the work of Constantino [15] for problems with sequence-independent setups for small bucket formulations with sequence-dependent setup times and costs. Idleness is not represented by an artificial product. However, the triangular setup conditions are assumed to hold. We will refer to Wolsey's model as the general small bucket model (GSB).

2.2. Big bucket approaches

In contrast to small bucket models, the planning horizon of a big bucket (BB) model is usually divided into longer periods of equal length. Time intervals in a BB model may represent a time slot of one week (or more) in practice [17]. In each period, multiple products can be manufactured. Releasing the "all-or-nothing" production principle of (most) SB models implies that a BB model includes the possibility to determine continuous lot-sizes.

The Capacitated Lot-Sizing Problem (CLSP) is a typical example of a big bucket model. It is closely related to the (small bucket) DLSP. Decision variables, parameters and objective function are comparable in both problems [17]. However, the CLSP does not include sequence-dependent setup costs and times. As a consequence, setup carry-over between period boundaries is not included either. Suerie and Stadler [59] use the simple plant location problem to obtain a tight new model formulation for setup carry-over in the CLSP with sequence-independent setup costs and times.

Sox and Gao [57] introduce the Generalised Capacitated Lot-sizing Problem (GCLP). The GCLP uses less binary variables for including setup carry-over in the CLSP with sequence-independent setup costs and no setup times. Sequence-independent setup times may be included; probably at the expense of additional computational effort. The authors also apply the network reformulation approach as proposed by Eppen and Martin [19] and compare the behaviour of a set of models. The results demonstrate that incorporating setup carry-over has a significant effect on both costs and lot-sizes.

We observe a tendency in simultaneous lot-sizing and scheduling to incorporate characteristics of small bucket models into

big bucket models. For confirmation we refer to proposed variants of the CLSP [1,2,28–30,33,34,43], variants of hybrid BB and SB models like the General Lot-sizing and Scheduling Problem (GLSP) [22,23,27,30,31,44,60], and variants of block planning approaches, originally introduced by Gunther et al. [32]. The literature review on extensions of capacitated lot-sizing by Quadt and Kuhn [50] and Guimarães et al. [31] confirm the trend in which (hybrid) BB approaches are preferred to SB models. For a complete overview we also refer to [11].

Including sequence-dependent setup costs and times for simultaneous lot-sizing and scheduling, is still a challenge.

2.3. Product decay

Although a vast body of literature exists on inventory management for perishable products, surprisingly little has been done to include product decay in traditional lot-sizing and scheduling models. Studies in which issues of integrating perishability are considered, mostly refer to managing perishability in production-distribution planning [10,4,5]. Amorim et al. [3] state that papers discussing simultaneous lot-sizing and scheduling for perishable goods are very rare.

In contrast to lot-sizing modelling, literature on inventory management has a long tradition of dealing with perishability. The early work of Van Zyl [61] and Nahmias [45] already focused on the concept of a fixed life time, where the product is lost after reaching a certain shelf life. Raafat [51] reviews literature on continuously deteriorating inventory models and discusses the concept of deterioration where a fixed percentage of the stored items is lost each period (exponential decay). The advantage is that those models do not have to keep track of the age of the stored items. We refer to Karaesmen et al. [38] and [6] for a more recent reviews on perishability in inventory systems.

One of the first contributions in the area of lot-sizing is provided by Soman et al. [55]. The paper studies shelf life considerations in the economic lot scheduling problem (ELSP). Models of this class usually assume constant demand, do not account for sequence-dependent setup times and aim to generate production cycles for several products on a single resource. Entrup et al. [18] propose three MILP models that integrate shelf-life issues into production planning and scheduling for an industrial case study of yoghurt production. The models use a block planning approach in which a block covers all products based on the same recipe. Shelf-life aspects are taken into account by considering a shelf-life-dependent pricing component that may also include inventory-holding costs. Lee and Yoon [42] consider a coordinated production-and-delivery scheduling problem that incorporates different inventory-holding costs between production and delivery stages. The results may be only applicable to limited situations but the study can be regarded as a first attempt to allow different (stage-dependent) inventory-holding costs. Chen et al. [10] conclude that papers discussing production scheduling and/or distribution of perishable goods are relatively rare. Amorim et al. [3] state that papers discussing simultaneous lot-sizing and scheduling for perishable goods are even rarer. The authors deal with simultaneous lot-sizing and scheduling of perishable products using a multi-objective framework to separate economic production tangible costs from intangible value of having fresher products in two conflicting objectives. A general overview on managing perishability in production-distribution planning is provided by [5]. We follow Entrup et al. [18] for modelling perishability by introducing an age-dependent pricing component where the value of the product decreases with the age (or increases in case of ripening).

For a general overview of lot-sizing problems we refer to several reviews of the past [17,39,41] and two more recent overviews [36,50].

3. Model formulation

This section presents two SB models and demonstrates the impact of (i) relaxing the triangular setup conditions and (ii) taking product decay into account. Section 3.1 describes the problem which is then modelled in Section 3.2 as a lot-sizing and scheduling problem with non-triangular setups. In Section 3.3 we extend the model such that it addresses product decay of inventory by including an age-dependent component in the inventory-holdings costs.

3.1. Outline of the lot-sizing and scheduling problem

The complete problem under study is characterised as follows:

- Consider N products to be scheduled over a finite planning horizon of T periods. Consider a small bucket problem i.e. in each time period at most one item can be produced at full capacity, or there is no production at all.
- A lot or batch of item i is defined as an uninterrupted sequence of periods in which production takes place for item i .
- Manufacturing items requires a common equipment or resource with limited capacity. Without loss of generality, machine capacity is normalised to 1 unit per period.
- Demand is assumed to be varying and deterministic and expressed in the number of required production periods.
- Demand must be satisfied (without backlogging) either by production in the same period or from stock. In the latter case, an inventory carrying cost is incurred.
- It is assumed that the Wagner-Whitin cost condition holds, i.e. given the setups, it always pays to produce as late as possible. This condition is also referred to as the absence of speculative motives for early production.
- Unit production costs are assumed to be constant over periods and are therefore ignored.
- A changeover between items incurs a loss of production capacity (non-zero setup times) and associated setup and changeover costs. Both are sequence-dependent.
- Setup costs are assumed to be proportional to changeover times.
- The triangular setup conditions refer to the changeover matrix A (with respect to setup times), the changeover matrix S (with respect to setup costs), or both:

$$a_{ij} \leq a_{ik} + a_{kj} \quad \text{and/or} \quad s_{ij} \leq s_{ik} + s_{kj} \quad \text{for all items } i, j, k \quad (1)$$

- If changeover and idleness occur in subsequent time intervals, we follow common practice and assume that periods of idleness are preceded by changeover time.
- In contrast to the DLSP, the setup state of the machine should be preserved over idle time. This assumption is made in many lot-sizing and scheduling models [17]. Generally speaking, it is also valid in FPI.
- Starting inventory levels for product i are assumed to be zero, i.e. $I_{i,0} = 0$ for all i .
- The objective is to determine the production sequence and lot-sizes that minimise the sum of setup and inventory carrying costs over the complete planning horizon.

3.2. Notation and model formulation

The following notation is used to formulate the problem: Parameters

N	number of items or products i.e. $i, j = 1 \dots N$
T	number of time intervals (i.e. periods) in the planning horizon; $t = 1 \dots T$

h_i	unit storage costs of item i at the end of a period
S_i	fixed setup costs for product i
$d_{i,t}$	demand for item i in period t (expressed in units of required capacity)
a_{ij}	changeover time between products i and j in units of lost capacity

Variables

$I_{i,t}$ inventory level of item i at the end of period t

$$Y_{i,t} = \begin{cases} 1 & \text{if product } i \text{ is produced in period } t \\ 0 & \text{otherwise} \end{cases}$$

$$V_{j,t} = \begin{cases} 1 & \text{if the machine is changing over to product } j \text{ in period } t \\ 0 & \text{otherwise} \end{cases}$$

$$W_{j,t} = \begin{cases} 1 & \text{if the machine is idle in period } t \\ & \text{and the next item to produce is product } j \\ 0 & \text{otherwise} \end{cases}$$

Now, the problem can be formulated as follows:

$$\text{Min} \left\{ \sum_j S_j \sum_t V_{j,t} + \sum_i \sum_t h_i I_{i,t} \right\} \quad (2)$$

s.t.

$$I_{i,t-1} + Y_{i,t} - d_{i,t} = I_{i,t} \text{ for all } i, t \quad (3)$$

$$\sum_i Y_{i,t} + \sum_j V_{j,t} + \sum_j W_{j,t} = 1 \text{ for all } t \quad (4)$$

$$Y_{i,t} - \sum_{k:k \neq i \neq j} \sum_{s=t+1}^{t+\tau-1} Y_{k,s} + Y_{j,t+\tau} \leq 1$$

for all $(i, j) \in A: = \{ (i, j) | a_{ij} > 0 \}$ and

$$t = 1, \dots, T - \tau \quad \tau = 1, \dots, a_{ij}$$

$$V_{j,(t-a_{i,j}+\tau)} \geq Y_{i,(t-a_{i,j}-1)} + Y_{j,t} + W_{j,t} - \sum_{k:k \neq i \neq j} \sum_{l=t-a_{ij}}^{t-1} Y_{k,l} - 1$$

for all $(i, j) \in A: = \{ (i, j) | a_{ij} > 0 \}$ and

$$t = (a_{ij} + 2), \dots, T \quad \tau = 0, \dots, (a_{ij} - 1)$$

$$\sum_j W_{j,t} + \sum_j V_{j,t+1} \leq 1 \text{ for } t = 1, \dots, T - 1 \quad (7)$$

$$W_{i,t} + \sum_{j \neq i} Y_{j,t+1} + \sum_{j \neq i} W_{j,t+1} \leq 1 \text{ for all } i, t = 1, \dots, T - 1 \quad (8)$$

$$Y_{i,t}, V_{j,t} \in \{0, 1\} \text{ for all } i, j, t \quad (9)$$

$$I_{i,t}, W_{j,t} \geq 0 \text{ for all } i, j, t \quad (10)$$

Objective function (2) minimises the sum of changeover and inventory-holding costs. Constraint (3) represents the inventory balance equations and assures demand $d_{i,t}$ for item i in period t is fulfilled without backlogging. Eq. (4), together with (9) and (10), guarantee that in each time interval the machine is either producing item i at full capacity ($\sum_i Y_{i,t} = 1$), changing over ($\sum_j V_{j,t} = 1$),

or idle ($\sum_j W_{j,t} = 1$) before manufacturing the next batch of an item. Constraint (5) assures that between two subsequent production batches i and j , sufficient time (a_{ij}) is reserved for changeover. For positive change over time ($a_{ij} > 0$), inequality (6) enforces the setup variables to be non-zero between two subsequent batches i and j ; if item j is produced in period t ($Y_{j,t} = 1$) or the machine is idle in period t ($W_{j,t} = 1$) before manufacturing item j in period t' ($t' > t$), setup variables $V_{j,(t-a_{i,j}+\tau)}$ should get a value of one for $\tau = 0, \dots, (a_{ij} - 1)$. The term $\sum_{k:k \neq i \neq j} \sum_{l=t-a_{ij}}^{t-1} Y_{k,l}$ in (6) represents the production of another item than i and j within time interval $[t - a_{ij}, t - 1]$. Constraints (7) assure that a setup cannot take place after idleness. Hence, a possible idle period will take place after a setup is counted for. Inequalities (8) prevent that idleness in period t before manufacturing item i is followed by the production (or preliminary idleness) of another item j . Finally, restrictions (9) and (10) define the integrality and non-negativity requirements.

3.3. Modelling product decay

When a traditional objective function like (2) is used for perishable items, traditional linear holding costs in lot-sizing models may disrupt a crucial balance between changeover costs on the one hand and inventory-holding costs on the other hand. Product decay has an impact on the remaining shelf life of products. This aspect is included by an age-dependent component in the inventory-holding costs [18].

Product decay of inventory can be incorporated in a SB type model like (2–10) in the following way. Let additional parameter $pr_i \geq 1$ represent the perishability rate of item i . Next, we redefine the inventory variables $I_{i,t}$ by $I_{i,t,q} \geq 0$ (for all i, t, q in which $q \leq t$) as the inventory level of item i at the end of period t , produced in period q . Now, objective function (2) is replaced by (2b) in which $(t-q)$ represents product age:

$$\text{Min} \left\{ \sum_j S_j \sum_t V_{j,t} + \sum_i \sum_t \sum_{q=1}^t h_i pr_i^{(t-q)} I_{i,t,q} \right\} \quad (2b)$$

Note that if the perishability rate is $pr_i = 1$ for all items i , then objectives (2) and (2b) are equal. Replacing (3) by (3a–3e) describes the age dynamics of the inventory levels:

$$I_{i,1,1} = Y_{i,1} - d_{i,1} \text{ for all } i \quad (3a)$$

$$I_{i,t,1} \geq I_{i,t-1,1} - d_{i,t} \text{ for all } i, t = 2 \dots T \quad (3b)$$

$$I_{i,2,2} = \sum_{\tau=1}^2 (Y_{i,\tau} - d_{i,\tau}) - I_{i,2,1} \text{ for all } i \quad (3c)$$

$$I_{i,t,q} \geq \sum_{\tau=1}^q I_{i,t-1,\tau} - d_{i,\tau} - \sum_{\tau=1}^{q-1} I_{i,t,\tau}$$

for all $i, t = 3 \dots T, q = 2 \dots t - 1$ (3d)

$$I_{i,t,q} = \sum_{\tau=1}^t (Y_{i,\tau} - d_{i,\tau}) - \sum_{\tau=1}^{t-1} I_{i,t,\tau} \text{ for all } i, t = 3 \dots T, q = t \quad (3e)$$

4. Numerical illustrations and benchmark

This section illustrates the impact of (i) (relaxing) the triangular setup conditions and (ii) incorporating product decay of inventory on optimal production schedules. We use small numerical examples

Table 1

Demand.

i/t	1	2	3	4	5	6	7	8	9	10
1						1			1	
2										1
3					1					2

to compare the behaviour and characteristics of the model with the general small bucket model GSB from literature [62]. Model formulation (2–10) is referred to as SB1, and formulation (2b), (3a–3e), (4–10) is referred to as SB2. We first consider a tailored problem for $N=3$ and $T=10$.

4.1. Impact of (non-)triangular setup and product decay

The impact of (relaxing) the triangular setup conditions and modelling product decay by age-dependent holding costs is demonstrated by three illustrative problem instances.

Example 1. Triangular setups, no product decay.

Consider inventory-holding costs of $(h_i) = (30, 25, 23)$. The (nonzero) values of demand $d_{i,t}$ and changeover times a_{ij} (expressed in required periods of capacity) are given by: Table 1 and 2.

Let the setup costs $S_i = 100$ and the processing time be unitary for all items i . Note that the triangular setup conditions (1) hold for matrix A_1 . However, the matrix is asymmetric. Table 3 shows the optimal production schedule for this instance.

Model formulations GSB, SB1, and SB2 with $pr_i = 1$ all obtain the same solution in Table 3 with total costs of 767 units. The setup state of the machine is preserved over the idle time in period 7.

Example 2. Non-triangular setup, no product decay.

Increasing the changeover time $a_{3,1}$ between item 3 and 1 to

$a_{3,1} > 2$ results into changeover matrix $A_2 = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & 2 \\ 3 & 1 & 0 \end{pmatrix}$ for which

triangular setup conditions (1) do not hold, as $a_{3,1} > a_{3,2} + a_{2,1}$. The optimal production schedule obtained by model SB1 and SB2 with $pr_i = 1$ is presented in Table 4. Obviously, the optimal production schedules in the Tables 3 and 4 are substantially different.

Model formulation GSB again produces the production schedule of Table 3. Clearly, this solution is infeasible; a changeover from item $i=3$ (in period 3) to item $j=1$ (in period 6) requires 3 time intervals. Apparently, using model formulation GSB for cases in which the triangular setup conditions (1) do not hold, implies that setup state changes will occur (a changeover from item $i=3$ to item $j=2$ in period 4 and from item $i=2$ to item $j=1$ in period 5) without a production change, i.e. no associated production for item 2.

Example 3. Non-triangular setup and product decay.

We include product decay of inventory by using age-dependent holding costs: a perishability rate $pr_i = 1.12$ is used for all items and applied to model SB2. All other data from Example 2 remain unchanged, including changeover matrix A_2 .

Model SB2 yields a completely different production schedule as

Table 2Changeover matrix A_1 .

i/j	1	2	3
1	0	1	2
2	1	0	2
3	2	1	0

Table 3

Optimal solution Example 1 total costs 767.

t	State	$\sum_j S_j V_{j,t}$	$\sum_i h_i I_{i,t}$
1	3		23
2	3		46
3	3		69
4	Setup	100	69
5	Setup	100	46
6	1		46
7	Idle		46
8	1		76
9	Setup	100	46
10	2		
Costs:		300	467

Table 4

Optimal solution Example 2 total costs 775.

t	State	$\sum_j S_j V_{j,t}$	$\sum_i h_i I_{i,t}$
1	1		30
2	1		60
3	Setup	100	60
4	Setup	100	60
5	3		60
6	Idle		30
7	3		53
8	3		76
9	Setup	100	46
10	2		
Costs:		300	475

presented in Table 5. Apparently, a small change in the balance between inventory holding- and changeover costs has a major impact on the generated production schedules, see Tables 4 and 5.

4.2. Complexity discussion

We analyse the complexity of formulations GSB, SB1 and SB2. Table 6 provides a general overview. The general problem size is expressed in a common notation, i.e. the dominating term of the number of constraints and variables. Table 6 shows that adding the functionality to cope with non-triangular setups for simultaneous lot-sizing and scheduling requires more constraints, see column “Constraints” for the rows GSB and SB1. On the other hand, formulation SB1 requires substantially less (binary) variables than model GSB. Nevertheless, the models GSB, SB1, and SB2 are potentially very large formulations. The impact of the time horizon on problem sizes for an SB-approach becomes most obvious for model SB2 (adding product decay).

Complexity considerations for (variants of) the DLSP are published in Bruggemann and Jahnke [8] and Salomon et al. [52]. If

Table 5

Optimal Solution Example 3 by model SB2 Total costs 916.38.

t	State	$\sum_j S_j V_{j,t}$	$\sum_i \sum_q h_i pr_i^{t-q} I_{i,t,q}$
1	3		23.00
2	Setup	100	25.76
3	2		53.85
4	Setup	100	60.31
5	1		61.36
6	1		65.12
7	Setup	100	72.94
8	Setup	100	81.69
9	3		72.35
10	3		
Costs:		400	516.38

Table 6
Complexity of the models.

Model	Constraints	Variables
GSB	$O[N^2T]$	$O[N^2T]$
SB1	$O[\lambda(A)N^2T]^a$	$O[NT]$
SB2	$O[NT^2]$	$O[NT^2]$

^a $\lambda(A) > 1$ depends on the values of the elements in changeover matrix A

setup times are ignored, it is rather easy to test whether a feasible solution exists by comparing cumulative demands (expressed in required capacity) with cumulative capacity. However, if setup times are considered, even the feasibility problem of DLSP is NP-complete [52]. Brueggemann and Jahnke (2000) prove that the DLSP is NP-hard in the strong sense. It is easy to show that the SB1 formulation encloses the general DLSP formulation of Brueggemann and Jahnke (2000) by making the setup times independent of the following product and by scaling the production speed to one unit per period. If the DLSP is NP-hard, then the complexity of the SB1 should at least include the complexity of the DLSP which makes the SB1 at least NP hard in the strong sense. The latter indicates the requirement of effective and efficient approximation techniques to generate good feasible solutions for larger problem dimensions. Nevertheless, the availability of a correct model formulation offers a unique possibility to measure the quality of heuristically generated solutions for small to medium sized examples.

4.3. Heuristic approach

In a small bucket model like SB2, the number of time periods T has a significant impact on the problem dimension and the computational complexity of the problem. Therefore, time-based decomposition approaches seem attractive. The concept of the relax-and-fix (R&F) heuristic [16,58] is a typical example of a time-oriented decomposition approach. The R&F belongs to the class of mathematical programming-based heuristics that use a mathematical programming (MP) procedure to generate solutions. Heuristics of this class have the advantage that modules of many commercial solvers can be used, possibly with some (minor) adaptation. Moreover, the R&F heuristic provides guidance for the assessment of the quality of generated solutions by generating both a lower and an upper bound on the optimal objective function value. This section describes a basic variant of an R&F heuristic [47] which is used in the numerical analysis of Section 5.

The relax-and-fix algorithm sequentially solves P different mixed integer programming problems, denoted by MIP_p with $1 \leq p \leq P$. The set of integer variables Q is partitioned into P subsets $Q_1 \dots Q_P$, $p=1, \dots, P$ corresponding to a partitioning of the time horizon. For model SB2, set Q_1 consists of the binary production and idle variables $Y_{j,t}$ and $W_{j,t}$ of the time periods $t=1, \dots, t_1$, Q_2 contains all binary variables of periods $t=t_1+1, \dots, t_2$ etc. up to Q_P consisting of all binary variables associated with the periods $t=t_{P-1}+1, \dots, T$. In each iteration k , the variables of subset Q_k are defined binary while all other variables in Q are either relaxed i.e. defined as continuous variables or fixed to their (binary) value found in earlier iterations.

We apply a backward procedure, i.e. in the first iteration subproblem MIP_P is solved in which the integrality restrictions are imposed on the variables in subset Q_P . In other words, production and idle decisions are only made within the window $t=t_{P-1}+1, \dots, T$. In the next iteration, the integer variables in the subset Q_P are fixed at their optimal values as found in iteration P . Next, problem MIP_{P-1} is solved to find integer values for the subset of binary variables in Q_{P-1} and so on. In each problem MIP_p the production and idle variables are fixed at their optimal values in earlier iterations.

We do not apply a common forward procedure in the R&F algorithm. As demand matrices for small bucket models are usually sparse, finding a feasible solution at the end of the R&F algorithm strongly depends on the solution procedure i.e. forward or backward. In a forward procedure, production is postponed in early iterations. If capacities are tight, the concept of fixing the production and idle variables at their optimal values from previous iterations, may easily lead to infeasible solutions if a forward procedure is applied. It should be mentioned that (only) problem MIP_P in the first iteration of the R&F algorithm is a relaxation of the original problem. This automatically implies that the objective value of problem MIP_P in the first iteration of the R&F procedure provides a valid lower bound on the optimal objective function value.

5. Numerical analysis

We investigate the computational behaviour of model SB2 and the potential value of a basic MP-based heuristic, i.e. a backward Relax-and-Fix solution procedure of Section 4.3. We compare the objective values reached and the required computing time to run the solution procedures. All problems were solved using Xpress-Mosel on a personal computer with an Intel Core i3 CPU, 2.13 GHz, RAM 4GB.

Test cases were generated from small to moderate size. Both the number of items N and the number of time periods T are varied according to $N = 3, 4, 5, 6$ and $T = 30, 40, 50, 60, 90, 120$. The perishability rate is set at $pr = 1.1$ and setup-costs $S_i = 500$ for all products i in all cases. The inventory-holding costs differ between items i but remain unaltered between cases. Demand is expressed in discrete units of production capacity and may occur at any timeslot in the planning horizon. Demand on a timeslot may be larger than a single unit of production capacity. The changeover matrix is asymmetric, sequence-dependent and such that the triangular setup conditions (1) do not hold. All changeover times $a_{ij} > 0$ for $i \neq j$.

Table 7 shows the results for a set of instances solved by model SB2. The first column indicates the problem size, i.e. the number of items N and the planning horizon T . Columns 2 and 3 refer to the objective function value of the optimal solution (Objv) and the time needed to find the solution (CPU). The results in column 3 confirm that finding optimal solutions requires high computation times for medium size instances. The search procedure for optimal solutions was interrupted after eight hours of CPU time, provided that for each value of $N = 3, 4, 5, 6$ at least three test cases were solved to optimality.

An interrupted solution procedure is indicated by entry ‘-’ in Table 7. Columns 4 to 6 refer to solutions found by the Relax-and-Fix (R&F) heuristic, i.e. the objective function value found in the final iteration (Objv_RF), the relative deviation between the optimal objective value and the objective value of the R&F solution (GAP), and the computation time needed to find the R&F solution (CPU_RF). For all instances in Table 7, the R&F construction heuristic performed remarkably well: for 10 out of 18 cases the optimal solution was found, and time savings were substantial.

6. Concluding remarks

This paper studies how to include realistic features of food processing industry (FPI) in models for simultaneous lot-sizing and scheduling, in particular (i) sequence-dependent setups (including non-triangular setups), and (ii) product decay of inventory due to perishability.

Table 7
Computational results.

N / T	Optimal solutions		Relax-and-fix		
	Objv	CPU	Objv_RF	GAP	CPU_RF
3/20	3475.7	2	3475.7	0.0%	1
3/30	5453.2	4	5453.2	0.0%	2
3/40	6805.6	17	6805.6	0.0%	6
3/50	8245.7	132	8245.7	0.0%	14
3/60	10341.2	12,371	10341.2	0.0%	74
3/90	–	–	15686.1	–	102
3/120	–	–	19852.7	–	115
4/20	3948.7	1	Infeasible	–	–
4/30	6114.9	5	6287.3	2.8%	5
4/40	8554.5	75	8554.5	0.0%	194
4/50	11373.5	268	11373.5	0.0%	33
4/60	14095.2	2380	14095.2	0.0%	453
4/90	–	–	20518.0	–	84
4/120	–	–	27595.2	–	407
5/20	3781.8	2	Infeasible	–	–
5/30	5996.5	51	6219.7	3.7%	15
5/40	9569.3	1821	10492.8	9.7%	284
5/50	12780.5	3405	13152.7	2.9%	51
5/60	14479.5	21,299	14888.0	2.8%	112
5/90	–	–	26143.3	–	193
5/120	–	–	36985.8	–	486
6/30	8322.8	970	8322.8	0.0%	41
6/40	10197.0	6565	10228.6	0.3%	135
6/50	13478.0	209,716	13478.0	0.0%	631
6/60	–	–	17848.2	–	1387
6/90	–	–	30353.3	–	770
6/120	–	–	39951.2	–	610

Although big bucket (BB) models are usually associated with medium term planning horizons, various extensions of these models are proposed for short term planning horizons too. From a computational point of view it is explainable to prefer models with large time intervals over small bucket (SB) approaches. However, we state that segmentation of the planning horizon is a key issue for simultaneous lot-sizing and scheduling, particularly in food processing industry. Using large time periods implies that some basic principles for lot-sizing and scheduling (unfoundedly) disappear from sight. We give two reasons to underpin the latter statement.

Firstly, if the objective for simultaneous lot-sizing and scheduling should include the best compromise between total setup costs and total inventory holding costs, a time oriented aggregation (e.g. in BB models and its variants) may easily disrupt the general principle of optimality for lot-sizing [48]. If time intervals represent long periods (e.g. a week or more) multiple batches can be produced in each period. Consequently, inventory costs for batches manufactured at the start of lengthy periods are assumed to be equal to inventory costs of lot sizes produced at the end of the same period. As a result, total inventory costs in (2) or (2b) are underestimated, the main principle of optimality for lot-sizing may become disrupted, and production schedules will change accordingly. Secondly, product decay in food processing industry is primarily associated with the “age” of products. Incorporating issues of perishability like product decay requires capturing the precise moments of production for manufactured products.

We developed an SB model that (i) can handle sequence-dependent setups (including non-triangular setups), (ii) addresses product decay by incorporating age-dependent holding costs. Small-scale examples are used to demonstrate the impact of non-triangular setups and product decay on the generated solutions. The models show how a small change in the balance between inventory holding- and changeover costs may generate significantly different solutions, especially when the triangular setup conditions do not hold.

As expected, the computational effort for the SB model is substantial. We performed exploratory research with a straightforward implementation of a Relax-and-Fix construction heuristic. Numerical tests show that the quality of R&F solutions is promising at manageable computational effort. Additional research is needed to extend the construction heuristic with improvement steps and to find more enhanced variants of the R&F and an effective (algorithmic) segmentation of the time horizon, e.g. by exploiting the sparse demand matrix in a SB approach. Smoothing the heuristic solution by creating some overlap between successive iterations may be an option for further research [48]. We refer to Escudero and Salmeron [20], Federgruen et al. [21], James and Almada-Lobo [35], Guimarães et al. [30] and Oliveira et al. [46] for an overview of various (hybrid) strategies in an R&F framework.

Another interesting question for further research is to improve computational performance by adding valid inequalities (VI's) a priori to the initial formulation. Preliminary research revealed that adding the inequalities $Y_{j,t} + W_{j,t} \leq Y_{j,t-1} + W_{j,t-1} + V_{j,t-1} \quad \forall j, t$ tightened the linear relaxation and provided substantially better heuristic solutions in an R&F framework. The inequalities are based on feasibility conditions within any production scheme and are added before calling the MP-solver.

The proposed model offers a unique possibility to measure the quality and performance of any (heuristic) approach.

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